

TECH-ARCH-1480

Architecture Reference Document

Metadata

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* Department: Architecture

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1. Document Scope and Executive Summary

The Architecture Reference Document (ARD) outlines the policies, procedures, and best practices for the Architecture department.

The ARD covers topics such as architecture principles, design patterns, and guidelines for developing modular, scalable, and maintainable systems.

2. Core Policies and Standard Operating Procedures

2.1 Architecture Principles

The Architecture department is committed to designing software applications that are:

- * Modular: Encapsulating business logic and minimizing tight coupling between components.
- * Scalable: Capable of handling increased traffic, data growth, or user demand without significant performance degradation.
- * Maintainable: Designed with maintainability in mind, featuring clear code organization, minimal dependencies, and easy debugging.
- * Secure: Implementing robust security controls to protect against unauthorized access, tampering, and data breaches.

2.2 Design Patterns

The Architecture department adopts the following design patterns:

- * Microservices-based architecture for developing distributed systems.
- * Event-driven architecture for handling asynchronous workflows and real-time processing.
- * Service-oriented architecture (SOA) for encapsulating business logic and providing a layer of abstraction between components.

2.3 Coding Standards

Architects are expected to adhere to the following coding standards:

- * Programming languages: Java, Python, C#, and JavaScript.
- * Code organization: Following the package structure, naming conventions, and commenting guidelines outlined in the company's coding standards.
- * Version control: Utilizing Git for source code management and collaboration.

2.4 Collaboration and Communication

Architects are expected to:

- * Collaborate with cross-functional teams, including development, testing, and operations teams.
- * Communicate design decisions and trade-offs clearly and effectively to stakeholders.
- * Participate in regular design reviews and feedback sessions.

3. Compliance Monitoring, Penalties, and Operational Governance

3.1 Compliance Monitoring

The Architecture department will:

- * Conduct quarterly architecture review meetings to ensure compliance with this document's guidelines and principles.
- * Monitor architecture design decisions against the company's technology roadmap and strategic objectives.

3.2 Penalties for Non-Compliance

Failure to comply with the policies and procedures outlined in this document may result in:

- * Verbal warnings or written reprimands for minor infractions.
- * Suspension or termination of employment for repeated or severe non-compliance.

3.3 Operational Governance

The Architecture department will:

- * Establish a governance model for architecture design decisions, including approval processes and decision-making authorities
- * Develop a risk management framework to identify, assess, and mitigate potential risks associated with architecture designs.

By following the guidelines and principles outlined in this document, the Architecture department can ensure the development of